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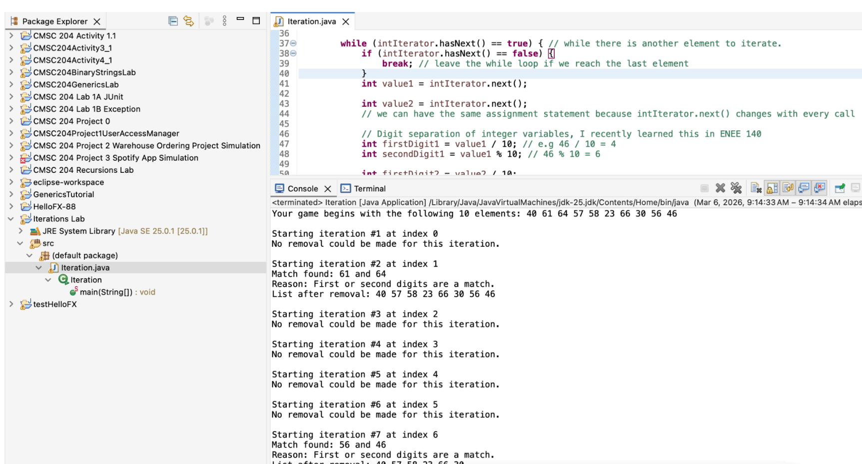
1. What did you learn?

I learned about iteration in Java and applied it to a real world context, in this case, a solitaire comparison game. I also learned about how we can potentially use iteration in conjunction with arrays to process and compare multiple different values for different criteria at once compared to standard array and ArrayList implementations which typically only process a single value/operation at once.

2. What challenges did you encounter, if any?

I initially struggled with getting the program to run and remove elements correctly but then I realized that the best way to achieve this is to directly remove the ArrayList object and use the ListIterator object to iterate. I also had to briefly review Random objects in Java since I have not used this feature in some time.

Screenshot of Test Run:



3. Convince me that your program (app) is working:

I will run through it manually here.
The game begins with the elements as shown.

Iteration 1:
40 and 61 have no matching digits, so the program moves on.

Iteration 2:
61 and 64 match in 1st digits, so they are both removed. The new list is 40 57 58 23 66 30 56 46

// Note: Since we are starting at 58 at this point, 57 and 58 cannot be removed.

Iteration 3:
58 and 23 have no matching digits, so the program moves on.

Iteration 4:
23 and 66 have no matching digits, so the program moves on.

Iteration 5:
66 and 30 have no matching digits, so the program moves on.

Iteration 6:
30 and 56 have no matching digits, so the program moves on.

Iteration 7:
56 and 46 match in 2nd digits, so they are both removed. The new list is 40 57 58 23 66 30

Since there are no more numbers to iterate, the program ends after displaying the final list again: 40 57 58 23 66 30